

Atreya Vedantam

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Theory of deep learning · feature learning · optimization · statistical learning

EDUCATION

IIT Madras

B.Tech., Electrical Engineering
2022 – present

CGPA 9.65/10

Department Rank 2 of ~175

HONORS

IUSSTF–Viterbi Scholar

Funded research program at USC,
2026

National Entrance Exams, 2022

JEE Advanced: All India Rank 315

JEE Main: All India Rank 221

(each of 1M+ candidates;

99.978th percentile)

KVPY Scholar

All India Rank 475, 2021;

99.05th percentile

Reviewer

NeurIPS 2026

LEADERSHIP

Head, Mathematics Club

IIT Madras, 2023–25. Led a team
of 70+. Founded the **Srinivasa**

Ramanujan Mathematics

Competition, India's first
olympiad-style contest for
engineering undergraduates: 600+
registrations from 100+
universities in year one.

Coordinator, AI Club

IIT Madras, 2024–25. Taught deep
learning to 1000+ students
nationwide; designed and ran an
AI/ML challenge for 350+
participants.

BEYOND RESEARCH

Novelist

Published a 60,000-word fantasy
novel, *The Tragedy of Girion*
(2018), begun in 2013.

Tennis

Inter-IIT contingent probable; gold
and bronze, inter-hostel
championship.

PUBLICATIONS & PREPRINTS

- *Characterization of Blind Code Rate Recovery in Linear Block Codes*. **IEEE ISIT 2026**.
- *Investigating Grokking in Logistic Regression*. arXiv 2602.08302; in preparation for AISTATS.
- *Intervention for Entanglement Routing in a Homogeneous Quantum Repeater Chain with Distillation*. In preparation.

RESEARCH EXPERIENCE

Feature Learning under Spectral Optimizers

May 2026 – present

IUSSTF–Viterbi Scholar, USC · Prof. Vatsal Sharan

- Generalizing the Neural Feature Ansatz to Muon and SpecGD; showed a whitened **SpecAGOP** tracks learned features where the standard AGOP fails, and scaling this to GPT-2-style transformers.
- Derived the AGOP as an optimal low-rank **compression** of the input; proposed a **covariance-weighted AGOP** for non-isotropic data.

Memory–Sample Trade-offs for Learning

May 2026 – present

Prof. Vatsal Sharan (USC) · Prof. Kabir Verchand (USC) · Prof. Ankit Pensia (CMU)

- Proved memory–sample **lower bounds** for *testing* planted-matrix streaming problems; the general theorem recovers and strengthens prior bounds (GKR'20).

Optimal Membership Inference in Linear Models

January 2026 – present

IIT Madras · Prof. Krishna Pillutla

- Characterizing optimal privacy attacks on ridge regression; derived per-point **susceptibility certificates** from residuals, leverage and effective dimension.

Grokking in Logistic Regression

Oct 2025 – present

Machine and Learning Theory Lab, IIT Madras · Prof. Chandrasekhar Lakshminarayanan

- Explained grokking through the **implicit bias** of gradient descent: late-time support-vector imbalance moves the bias long after the data is fit.
- Proved non-asymptotic **phase-separation** bounds; showed class imbalance is necessary.

Entanglement Distribution in Quantum Networks

Aug 2025 – present

B.Tech. Project, IIT Madras · Prof. Krishna Jagannathan

- Derived exact entanglement rates for quantum repeater chains with finite memory and decoherence; characterized when **distillation** outperforms swapping.

Blind Code Rate Recovery

Jan 2025 – present

Satlab, IIT Madras · Prof. Radha Krishna Ganti

- Proved **performance bounds** for rank-recovery heuristics under noise; validated on over-the-air LDPC captures.

Quantitative Research Intern

May – Jul 2025

Dynamic Technology Lab, Singapore

- Modeled mutual-fund positions; built a spectrogram-based co-integration metric for stock pairs.

RELEVANT COURSEWORK

Convex Optimization · Foundations of Deep Learning · Machine Learning · Information Theory · Probability · Stochastic Processes · Linear Algebra · Computability & Complexity